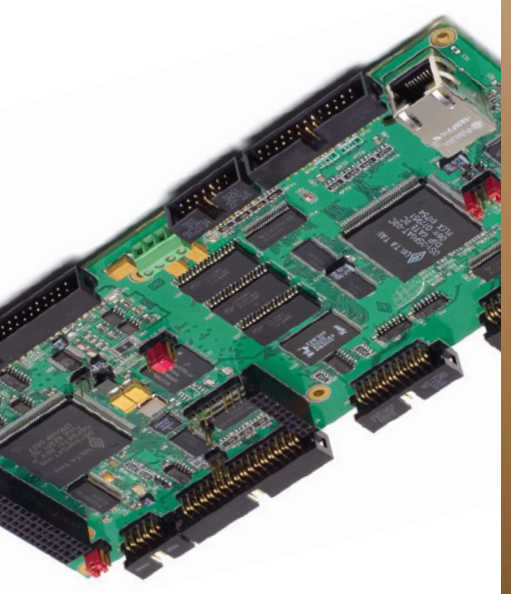
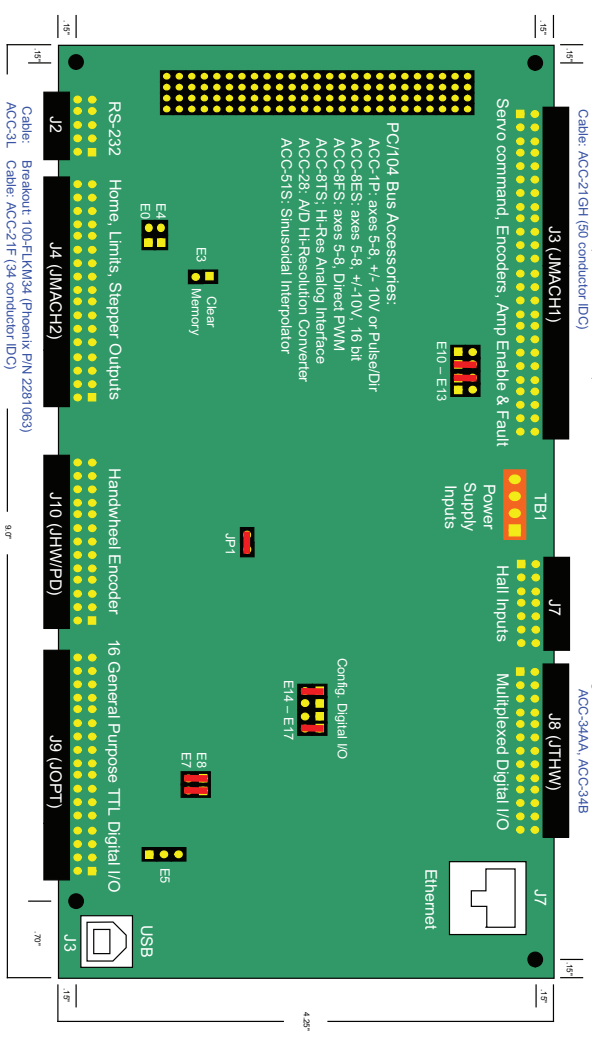
Quick Start Guide

Clipper
Turbo PMAC2 ETH-Lite

DELTA TAU



The **Clipper** is a complete 4 axis motion controller in a 9"x4.25" footprint. It provides +/- 10V & Pulse/Direction command outputs with a full complement of axis I/O (Home, Limits, Amp Enable & Fault). Communication is via Ethernet/USB/RS-232.



The **Clipper** accepts **Delta Tau's** 'stackable' PC/104 accessories:

- ACC-1P - 4 axes & optional 8/8 I/O channels
- ACC-8ES - 4 channels, 16 bit +/- 10V DAC outputs
- ACC-8FS - 4 channels of Digital Direct PWM
- ACC-8TS - Hi-Res Analog Interface
- ACC-51S - high resolution sine interpolator
- ACC-28B - A/D Hi-Resolution converter

Available cable / breakout accessories:

- Cables: 50 pin (ACC-21GH), 34 pin (ACC-21F), 9 pin (ACC-3L)
- Breakouts: 50 pin (100-FLKM50), 34pin (100-FLKM34)

Clipper Setup Overview

Step #1: Required materials (on **Clipper CD**):
- Clipper Hardware Reference Manual (**CHRM**)
- Turbo PMAC 2 User Manual (**TPUM**)
- Turbo PMAC 2 Software Reference Manual (**TSRM**)
- PMAC Executive Pro2 software (**PEWIN**)
- Motor Setup Files (**Clipper CD**)

Step #2: Jumper settings - the factory default settings are appropriate for most applications. For specific jumper descriptions please refer to the **E-Point Jumper Descriptions** in the **CHRM**.

Step #3: Connect Power Supplies - Please refer to the **Power Supplies** section of the **Machine Connections** chapter (**CHRM**) and/or the **Clipper Roadmap ***.

Step #4: Install **PEWIN** from the **CD** - This software provides you with the user interface, utilities, and drivers for setting up, exercising, and programming Clipper.

Step #5: Configure and Establish communication:
USB or RS-232: select **General Setup and Options** from the **PEWIN** Setup window.
Ethernet: please refer to the **Communications** section of the **Software Setup** chapter (**CHRM**).

Step #6: Connect Home and Limit sensors - Please refer to the **Overtravel and Home Switches** section of the **Machine Connections** chapter in the **CHRM** and/or the **Clipper Roadmap ***. You can test these devices / connections using **PEWIN's Motor Status** window.

Step #7: Connect Motor Signals - Please refer to the **Motor Signal Connections** section of the **Machine Connections** chapter in the **CHRM** and/or the **Clipper Roadmap ***.

Step #8: Define your motor setup variables - the CD includes example motor setup files for both servo and stepper motors. You can download these via **PEWIN**, or you can refer to the Basic Motor Setup section (the next 'panel' of this brochure).

Step #9: Auto-Tune your servo motors - From **PEWIN** select **PMAC Tuning Pro2** from the **Tools** menu.
A) From the **Position Loop** menu select **Open Loop Test**. The basic operation of the amplifier, motor, and encoder will be verified.
B) Select the AutoTune button. Select **Auto Select Bandwidth** and then 'click' the **Auto Tune** button. Follow the on-screen instructions.

Step #10: Create a backup of all controller settings - from the **PEWIN Backup** menu select **Upload Configuration**.

Your **Clipper** is now ready to start **Jogging**, **Homing**, and executing **Motion and PLC programs**.

* The Clipper Roadmap is on the backside of this page

Basic Motor Setup - define prior to moving

+/- 10V (Filtered 12 Bit DAC) Servo Setup Variables [1]

```
I7000 = 1001      ;PWM frequency 29.4kHz, PWM 1-4
I7001 = 5         ;Phase Clock 9.8kHz
I7002 = 3         ;Servo frequency 2.45kHz
I7003 = 1746      ;ADC frequency
I7100 = 1001      ;PWM frequency 29.4kHz, PWM 5-8
I7103 = 1746      ;ADC frequency
I70n6 = 0         ;Output mode: PWM
Ixx69 = 1001      ;DAC limit 10Vdc
I10 = 3421867     ;Servo interrupt time
```

Stepper (Pulse / Direction) Setup Variables [1]

Variables - Primary

```
Ix02 = $78004 *   ;Command Output Address (* motor #1)
I7m03 = 2258 *    ;Clock Control (*=default, 327 KHz step rate )
I7m04 = 15 *      ;Pulse Width (*=default, 1.526 usec.)
I7mn6 = 3         ;Output Mode - Pulse
I7mn7 = 2         ;Output Pulse Polarity (2 = low)
I7mn8 = 0         ;Direction Signal Polarity (+ = low)
Ixx69 = 2114      ;Pulse Frequency Limit (327 KHz)
                   Position Feedback (typical)
I7mn0 = 8         ;Open loop, Encoder Decode = Internal [2]
I7mn0 = 3 or 7    ;Closed loop, X4 CW, X4 CCW
```

General Motor Setup Variables

```
Ixx00 - Motor xx Activate
Ixx02 - Motor xx Command Output Address
Ixx03 - Position xx Loop Feedback Address
Ixx13, Ixx14 - Soft Position Limits
I7mn0 - Encoder n Decode Control
```

+/- 10V (16 Bit DAC) Servo Setup

Please refer to the Clipper Hardware Ref. Manual

Digital Direct PWM Setup

Please refer to the Clipper Hardware Ref. Manual

Software - Primary

PEWin32 Pro2
- View | Terminal
- View | Motor Status
- View | Motor Setup Summary
- View | Jog Ribbon

Software - Secondary

Turbo Setup Pro2 [3]
PMAC Tuning Pro2

[1] - Example files on **Clipper CD** (CD:\Motor Setup Files\)
[2]- Tune for best performance with "Tuning Pro2". 'Auto Tune' with 'Amplifier Type' = Velocity Loop
[3] - Refer to "Setup Pro2" notes (CD:\Documentation\Errata Notes.txt)

Jogging & Homing

Commands - Primary

Jogging: J/, J+, J-, J=, J:
Homing: Home (HM), HomeZ (HMZ) [1]

Variables - Primary

```
Ix19 - Motor xx Max. Permitted Jog/Home Accel
Ix20 - Motor xx Jog/Home Acceleration Time
Ix21 - Motor xx Jog/Home S-Curve Time
Ix22 - Motor xx Jog Speed
Ix23 - Motor xx Homing Speed and Direction
Ix24 - Motor xx Flag Address and Modes
Ix26 - Motor xx Home Offset
I7mn2 - Servo IC m Channel n Capture Control
I7mn3 - Servo IC m Channel n Capture Flag Select
```

Commands - Secondary

%{constant} - Feedrate Override

Variables - Secondary

```
Ix00 - Motor xx Activate
Ix03 - Position xx Loop Feedback Address
Ix04 - Velocity xx Loop Feedback Address
Ix13 - Motor xx Positive Soft Position Limit
Ix14 - Motor xx Negative Soft Position Limit
Ix19 - Motor x Max. Permitted Jog/Home Accel
Ix69 - Motor xx Output command limit
I7mn0 - Encoder n Decode Control
```

Status Bits - Primary

```
Motor activated - X:$000xB0,23,1
Open loop mode - X: :$000xB0,18,1
Negative end limit set - X:$000xB0,22,1
Positive end limit set - X::$000xB0,21,1
Stopped on Position Limit - Y:$000xC0,11,1
Home search in progress - X:$000xB0,10,1
Home Complete - Y:$000xC0,10,1
Amp fault error - Y:$000xC0,3,1
Fatal following error - Y:$000xC0,2,1
```

Status Bits - Secondary

```
Running a program - X:$000xB0,17,1
Desired velocity zero - X:$000xB0,13,1
Background in-position - Y:$000xC0,0,1
```

Software - Primary

PEWin32 Pro2
- View | Terminal
- View | Motor Status
- View | Motor Setup Summary
- View | Jog Ribbon

Software - Secondary

Turbo Setup Pro2 [2]
PMAC Tuning Pro2

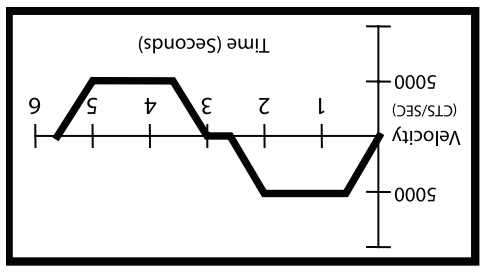
[1] - Example files on **Clipper CD** (CD:\Homing)
[2] - Refer to "Setup Pro2" notes (CD:\Documentation\Errata Notes.txt)

Motion Programs

```
;********** Set-up and Definitions **********
DEL GAT          ; Erase any defined gather buffer
&1               ; Coordinate System 1
CLOSE            ; Make sure all buffers are closed
#1->1000X        ; Motor 1 = X-axis, units = inches (1000 counts)

;********** Motion Program Text **********
OPEN PROG 1      ; Open buffer for program entry, Prog #1
CLEAR            ; Erase existing contents of buffer
LINEAR           ; Blended linear interpolation move mode
ABS              ; Absolute mode moves specified by position
TA500            ; Set 1/2 sec (500 msec) acceleration time
TS0              ; Set no S-curve acceleration time
F5               ; Set feedrate (speed) of 5 inches per sec
X10              ; Move X-axis to position 10"
DWELL500         ; Stay in position for 1/2 sec (500 msec)
X0               ; Move X-axis to position 0
CLOSE            ; Close buffer - end of program

;********** To run this program **********
&1 B1 R          ;Coordinate System 1 (&1), point to
                 ;Beginning of Program 1 (B1), Run (R)
```



Accessories

Axis Interface Boards

ACC-1P, Additional 4 axis interface (PC/104)
ACC-8ES, 4 Channel Analog Servo (16 bit) interface (PC/104)
ACC-8FS, 4 Channel Digital Direct PWM interface (PC/104)

Feedback Interface Boards

ACC-1P, Additional 4 axis interface (PC/104)
ACC-51S, High Resolution Sinusoidal Interpolator (PC/104)
ACC-8TS, Interface to ACC-28B Hi-Res. A/D inputs (PC/104)

Digital I/O boards

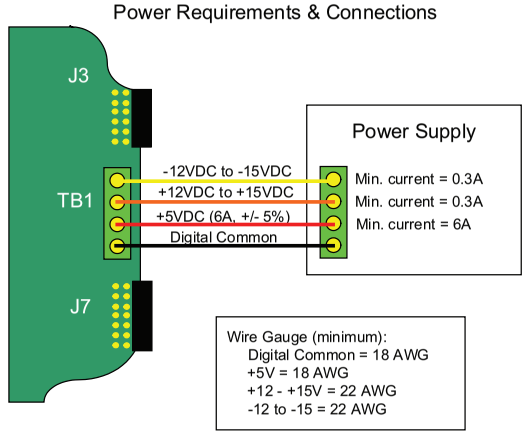
ACC-1P Opt 1, 8 inputs, 8 outputs (+5 - +24V) (PC/104)

Connector Accessories

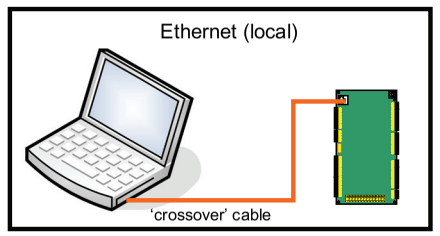
ACC-3P - 50 pin to 60 pin JMACH adapter board
FLKM50: terminal block, 50 contact

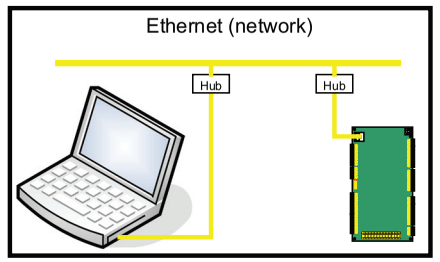
Clipper 'Roadmap'

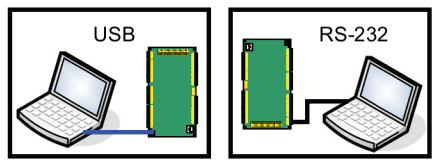
Power Requirements & Connections



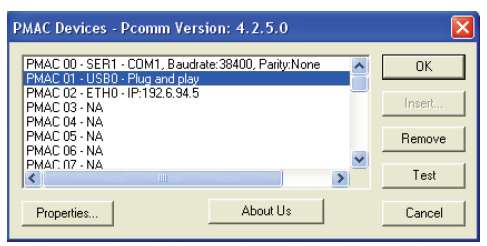
Available Communications



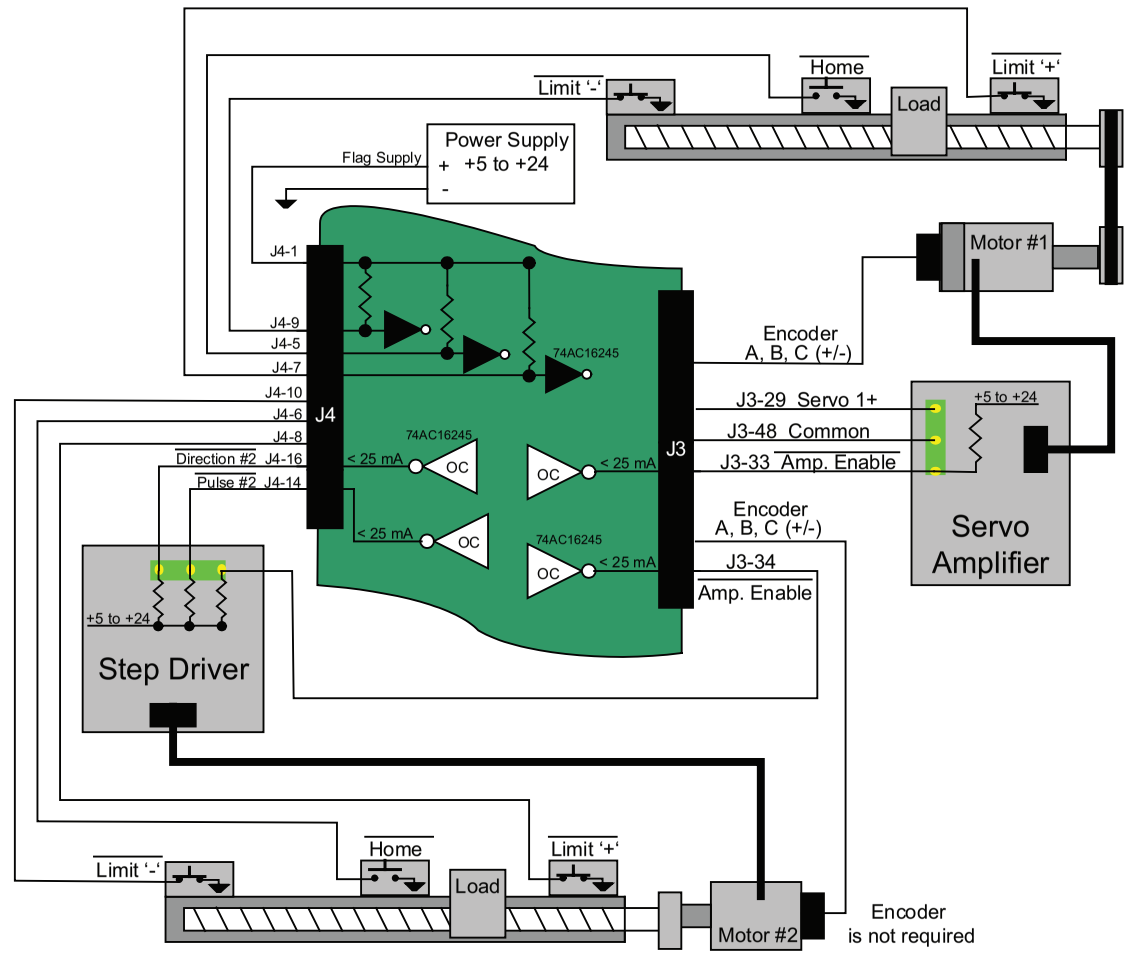




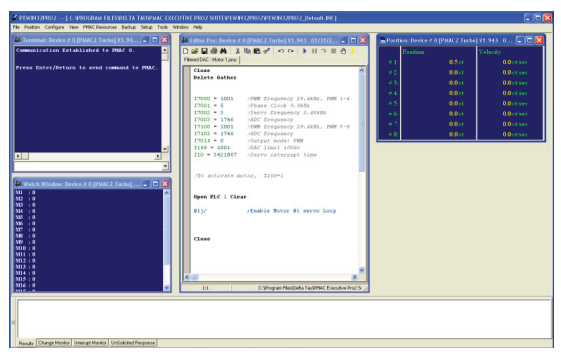
Select, configure, and test communications from the PEWIN **Setup** menu (**General Setup and Options**)



Typical Motor System Connections (example: 1 servo & 1 stepper)



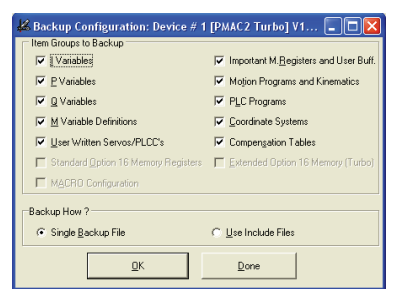
Use PEWIN to: create & download programs, monitor position, velocity, following error, I/O status.

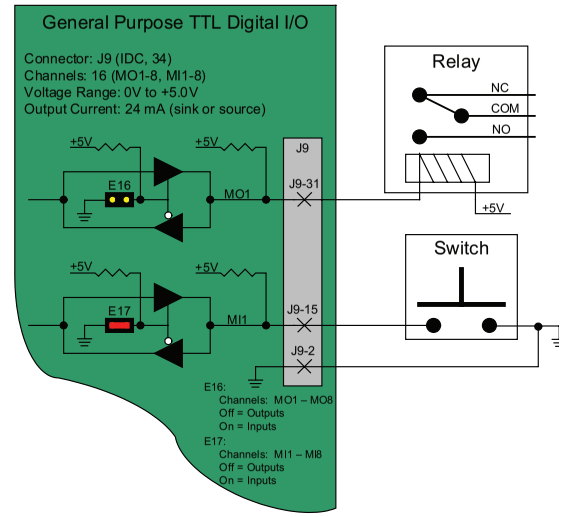


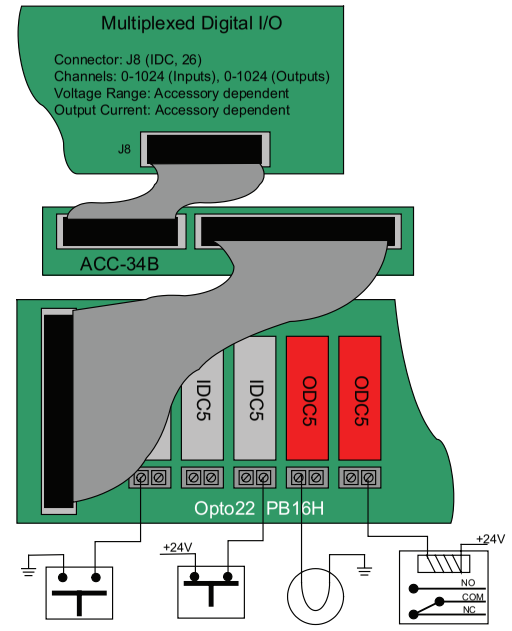
Display Motor Status (**PEWIN** | **View** | **Motor Status**)



Safeguard your data - from within PEWN you can backup all controller settings and programs by selecting **Upload Configuration** from the **Backup** menu.







Modbus/TCP Digital I/O

